

Back-to-School Crossword

Kevin Chen, Aaron Qin

1	2
3	

ONE ACROSS

MathEXplained High School is building a new student portal for the upcoming year. The editors-in-chief assign two student programmers, Justin and Kyle, to build the backend database.

Every day, Justin writes $\frac{1}{x}$ of the day's code, whereas Kyle writes $\frac{1}{y}$ of the day's code. When their work is put together, they write exactly $\frac{1}{900}$ of the total code required every day.

If Justin contributes at least as much work as Kyle every day, how many different integer pairs (x, y) can exist?

ONE DOWN

For the annual Back-to-School pep rally, the MathEXplained student council is painting a massive triangular school spirit pennant directly onto the gymnasium floor. The three sides of the pennant measure exactly 13 feet, 14 feet, and 15 feet.

To finish the design, the art committee wants to place a round spotlight at the pennant's circumcenter O and the Math-EXplained logo at the incenter I . The math club is tasked with calculating the exact distance between the spotlight and the emblem. To make the numbers easier to work with on their banner blueprint, they decide to compute the value of $64 \cdot OI^2$. What value do they get?

TWO DOWN

Elias takes the bus to school. Today, he noticed a regular octagonal STOP sign. This STOP sign is special: its vertices are colored using three colors so that adjacent vertices have different colors and all three colors are used. Two colorings are considered equivalent if one can be obtained from the other by a rotation. How many distinct colorings are there for the STOP sign?

THREE ACROSS

A high school art teacher, Mr. Vance, is preparing red, green, and blue paint for a back-to-school mural.

He stores the paint in three special containers. The volumes of paint in liters— a , b , and c —happen to be the three exact roots of the polynomial equation $x^3 - 3x + 1 = 0$.

For a mixing experiment, the students need to calculate the combined material strengths of these batches. Mr. Vance writes the following expression on the chalkboard and promises extra credit to anyone who can solve it before the bell rings:

$$(a^2 + b^2)(b^2 + c^2)(c^2 + a^2)$$

Can you solve the expression numerically to get extra credit?